Americas' Energy Dominance

US Shale, LNG Export Strategy, and Latin America's Hydrocarbon Wealth

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Executive Summary

The Americas have emerged as the decisive axis of global energy geopolitics. The United States now commands the world's largest liquefied natural gas export position, having overtaken both Australia and Qatar, while its tight oil machine — anchored in the Permian Basin — continues to generate the plurality of non-OPEC production increments. Brazil is simultaneously executing an offshore frontier expansion of hemispheric significance, with Petrobras preparing to drill its first wells on the Equatorial Margin in 2026. These twin dynamics — US hydrocarbon primacy and Brazilian deepwater ascent — are reshaping supply balances, trade routes, and geopolitical leverage across the Atlantic and Pacific basins. Against this backdrop, the US energy policy environment has shifted sharply toward domestic resource maximisation under the January 2025 National Energy Emergency declaration, creating a structural tension with the climate commitments encoded in the Inflation Reduction Act. Mexico's transition ambitions remain nascent and constrained. Canada's pipeline infrastructure faces persistent regulatory and security headwinds. This brief maps the four principal axes of Americas energy: US tight oil and LNG supremacy, Brazil's offshore frontier, the US policy inflection, and hemispheric energy security dynamics.

I. US Tight Oil and Permian Dominance: The Non-OPEC Engine

1. US tight crude output is the dominant driver of near- and medium-term non-Declaration of Cooperation (DoC) liquids supply growth globally. Even as growth rates moderate, annual increments in the medium term are projected to average 0.5 million barrels per day, accounting for more than 40 percent of total non-DoC liquids supply growth during this period. The Permian Basin remains the structural centre of gravity for US tight crude output. [RAG: OIL_ENERGY_RAG — OPEC World Oil Outlook 2024, p.169]

2. Across most projection cases in the EIA's Annual Energy Outlook 2026, US crude oil production is expected to fluctuate within a relatively narrow band: a modest decline in the early 2030s, a recovery around 2040, and a subsequent decline toward 2050. Prime acreage quality and continued technological advancement in resource recovery are identified as the key determinants of trajectory. [RAG: OIL_ENERGY_RAG — EIA Annual Energy Outlook 2026 Narrative, p.24]

3. Monthly US tight oil production data through 2025 show the Permian leading all formations by a substantial margin, with the Bakken, Eagle Ford, Niobrara-Codell, Austin Chalk, Mississippian, and Woodford formations constituting the secondary tier. This formation-level composition has remained structurally stable, with the Permian's share of total tight oil output continuing to widen. [RAG: OIL_ENERGY_RAG — EIA Short-Term Energy Outlook July 2026, p.29]

4. Advanced exploration techniques, frequently subsidised, drove a 9 percent increase in US gas reserves and a 12 percent increase in US oil reserves between 2017 and 2018, illustrating the resource

base's responsiveness to technology deployment and the long-run expandability of the unconventional resource endowment. [RAG: CLIMATE_RAG — IPCC AR6 Energy Systems Chapter 6]

5. The costs of extracting oil and gas globally have fallen substantially through hydraulic fracturing and directional drilling applied to unconventional reservoirs. Although unconventional extraction remains more expensive than conventional production, the large availability of unconventional resources has exerted sustained downward pressure on global prices, restructuring competitive dynamics across all producing regions. [RAG: CLIMATE_RAG — IPCC AR6 Energy Systems Chapter 6]

II. US LNG Supremacy: The New Global Gas Architecture

6. The United States has overtaken both Australia and Qatar to become the world's largest LNG exporter. International gas trade fluctuated between 900 and 1,000 billion cubic metres between 2017 and 2023, registering an overall fall of 2.7 percent to 936 bcm in 2023. Within this total, LNG exports rose 1.8 percent to 549 bcm. LNG now accounts for nearly 59 percent of all globally traded gas, a structural shift from pipeline dependency that disproportionately advantages the US given its shale-backed supply flexibility. [RAG: OIL_ENERGY_RAG — Energy Institute Statistical Review of World Energy 2024, p.46]

7. Global gas demand remained effectively stable in 2023, rising by just 1 bcm — insufficient to recover the 0.4 percent (15 bcm) decline recorded in 2022. North America dominates production in the Energy Institute's regional decomposition, with South and Central America constituting a secondary producing region whose gas market architecture is increasingly shaped by Petrobras pricing agreements and city-gate distribution dynamics. [RAG: OIL_ENERGY_RAG — Energy Institute Statistical Review of World Energy 2024, p.38]

8. The emergence of LNG markets has provided opportunities to export natural gas that were previously unavailable, fundamentally altering the geopolitics of gas dependency. Nations and trading blocs previously locked into pipeline relationships now have structural alternatives, and the US export position translates directly into diplomatic leverage — most consequentially in redirecting European supply away from Russian pipeline dependence following 2022. [RAG: CLIMATE_RAG — IPCC AR6 Energy Systems Chapter 6]

9. The US natural gas pipeline network, documented by the EIA's Gas Transportation Information System, constitutes the infrastructure backbone enabling LNG export growth. The network's routing decisions carry long-term commercial implications: the Rockies Express pipeline, for example, has documented operational history reflecting the complexities of matching midstream infrastructure with shifting production geographies. [RAG: LNG_ENERGY_RAG — Oxford Institute for Energy Studies NG195, p.62]

10. The US Department of Energy's December 2024 LNG export assessment — the most recent in a series dating to the EIA's 2012 study — evaluates the energy, economic, and environmental consequences of expanded US LNG exports. These assessments form the regulatory foundation for export authorisation decisions and shape the pace at which US LNG capacity can be licensed and financed for additional tranches of export growth. [RAG: LNG_ENERGY_RAG — DOE LNG Export Assessment December 2024, p.17]

11. The EIA's Short-Term Energy Outlook of July 2026 projects world liquid fuels consumption through 2027 across OECD and non-OECD blocs, with non-OECD growth continuing to outpace OECD demand. On the supply side, non-OPEC production growth is projected to lead global supply increments in the same horizon, reinforcing the Americas' structural role as the swing supply region for global oil and gas markets. [RAG: LNG_ENERGY_RAG — EIA Short-Term Energy Outlook July 2026, p.16]

III. Brazil's Offshore Frontier: The Equatorial Margin and the Pre-Salt Arc

12. Brazil is one of the largest economies in the world and an energy powerhouse, underpinned by its oil and natural gas reserves, abundant natural resources, and a well-developed energy industry. The country plays a significant role in international organisations focused on economic development and the energy sector, and its output trajectory makes it a tier-one variable in medium-term non-OPEC supply planning. [RAG: OIL_ENERGY_RAG — OPEC World Oil Outlook 2025, p.284]

13. Brazil is projected to be the second-largest source of non-DoC medium-term liquids supply growth. Total output is projected to rise materially through the medium term, driven primarily by Petrobras-led deepwater development. Crude and natural gas liquids constitute the dominant output categories in Brazil's supply profile, with biofuels adding a differentiated component that no other major oil producer replicates at comparable scale. [RAG: OIL_ENERGY_RAG — OPEC World Oil Outlook 2024, p.163]

14. The Equatorial Margin represents Brazil's newest and most strategically significant offshore frontier. Stretching over 2,200 kilometres from the far north of the state of Amapá to the coastline of Rio Grande do Norte, the region encompasses five basins previously assessed as prospective but commercially undeveloped. The scale of this frontier — and its geographic position bridging the Atlantic approaches — makes its development timeline a key variable for post-2030 global supply. [RAG: OIL_ENERGY_RAG — OPEC World Oil Outlook 2025, p.292]

15. Petrobras plans to begin drilling new wells in the Equatorial Margin in 2026, commencing with the Foz do Amazonas and Pará-Maranhão basins. The first well represents a defining commitment to the frontier and a significant escalation of Petrobras's exploration ambition beyond the established pre-salt Santos and Campos basin infrastructure. Successful appraisal would underpin the next phase of Brazil's oil production growth into the 2030s. [RAG: OIL_ENERGY_RAG — OPEC World Oil Outlook 2025, p.293]

16. Beyond offshore oil, Brazil remains well-positioned for wind power expansion, with consistently strong wind resources — particularly in the Northeast. As investment in renewable energy infrastructure accelerates, wind power is expected to play an increasingly central role in strengthening the country's energy security and advancing its decarbonisation commitments, providing a hedge against the long-run stranding risk inherent in hydrocarbon-dependent fiscal structures. [RAG: OIL_ENERGY_RAG — OPEC World Oil Outlook 2025, p.299]

17. Brazil's energy sector accounts for a significant share of the country's greenhouse gas emissions, with the oil and gas component representing 32 percent of the energy sector's emissions and 8 percent of Brazil's total GHG emissions in 2023. Carbon capture, utilisation, and storage (CCUS) is identified as a potential decarbonisation mechanism for the energy sector, with Bioenergy with CCS (BECCS) representing a carbon removal pathway that Brazil's bioeconomy is structurally well-suited to pursue. [RAG: OIL_ENERGY_RAG — OPEC World Oil Outlook 2025, p.303]

18. Brazil's city-gate natural gas pricing — governed by contracts between Petrobras and local gas distributors — is documented in the OIES Global Gas Demand 2025 report. This pricing architecture constrains the commercialisation of domestic gas relative to export alternatives, creating a structural tension between Petrobras's upstream economics and the downstream affordability requirements of Brazil's industrial consumers. [RAG: LNG_ENERGY_RAG — Oxford Institute for Energy Studies NG202, p.100]

19. OPEC's long-range production scenarios (EIA International Energy Outlook 2017 Reference Case) projected non-OPEC crude and lease condensate growth contributions from Russia, Canada, Brazil, and Kazakhstan through 2040, with Brazil singled out as a distinct contributor in the medium-to-long-run horizon. This framing has proved prescient: Brazil's pre-salt ramp-up since 2017 has consistently

outperformed early projections and is now a structural pillar of non-OPEC supply planning. [RAG: LNG_ENERGY_RAG — EIA International Energy Outlook 2017, p.22]

IV. US Energy Policy Inflection, Mexico's Transition, and Hemispheric Security Dynamics

20. The United States appears to be undergoing a fundamental shift in energy policy orientation, establishing domestic resource development as a national priority. The National Energy Emergency declaration of January 2025 frames energy security as a core national interest, signalling a strategic reorientation toward maximising domestic hydrocarbon output and accelerating permitting for fossil fuel infrastructure. [RAG: OIL_ENERGY_RAG — OPEC World Oil Outlook 2025, p.59]

21. The Inflation Reduction Act (IRA) is assessed in the UNEP Emissions Gap Report 2023-2024 as bringing the United States roughly two-thirds of the way toward meeting its Nationally Determined Contribution targets for 2030, with reductions of up to 1 GtCO2-equivalent over a no-policy scenario. However, the IRA also received criticism for provisions that allow expanded oil and gas exploration, creating a structural tension between its climate finance architecture and the resource development imperative embedded in the January 2025 declaration. [RAG: CLIMATE_RAG — UNEP Emissions Gap Report 2023-2024, Chunk 128]

22. In February 2023, Mexico announced the Sonora Plan — a renewables energy initiative aimed at deploying renewable energy and attracting low-carbon investment in the north of the country. The Sonora Plan represents a step toward accelerating Mexico's energy transition; however, it is assessed as the only new renewable energy project of scale announced in the relevant review period, indicating that Mexico's transition momentum remains narrow and structurally limited relative to its economic size and energy system complexity. [RAG: CLIMATE_RAG — UNEP Emissions Gap Report 2023-2024, Chunk 138]

23. Canadian energy firms have faced elevated and sustained threats from cyber and physical attacks, as documented by the Canadian Security Intelligence Service. The pipeline sector's vulnerability to both espionage and physical sabotage has been a recurring theme in national security assessments, with CSIS explicitly identifying pipeline infrastructure as a high-value target. The Energy East pipeline — which would have carried 1.1 million barrels per day of Alberta oil sands crude to Canada's east coast — had its National Energy Board hearings suspended in 2016 after protests disrupted proceedings, illustrating the regulatory and social fragility of large-scale North American pipeline infrastructure. CNA[2017-01-18]; CNA[2016-08-31]

24. The collapse of US shale economics in 2015-2016 triggered a cascading challenge to the pipeline sector: oil and gas producers attempted to use Chapter 11 bankruptcy protection to shed long-term pipeline contracts, placing the US$500 billion pipeline sector under direct financial stress. Two court disputes adjudicated whether upstream insolvency could invalidate midstream contracts — a systemic question with long-run implications for the bankability of pipeline infrastructure investment across the Americas. CNA[2016-02-23]

25. OPEC's Annual Statistical Bulletin 2025 presents crude oil production data for both OPEC and non-DoC members, including Mexico, which is listed as a non-DoC liquids producer alongside Russia, Kazakhstan, and other major producers outside the Declaration of Cooperation framework. Mexico's inclusion in this tier reflects its continued relevance as a producing state, even as Pemex's structural decline in output has reduced its weight in hemispheric supply planning relative to the US and Brazil. [RAG: OIL_ENERGY_RAG — OPEC Annual Statistical Bulletin 2025, p.31]

26. The stranded asset risk framework, as analysed in the IPCC AR6, has direct relevance to Americas energy geopolitics: approximately 30 percent of oil, 50 percent of gas, and 80 percent of coal reserves would remain unburnable if warming is limited to 2°C. For resource-dependent economies in the Americas — including Venezuela, Ecuador, and Colombia — this framing represents a long-run fiscal and sovereign risk that shapes both the urgency of diversification and the political economy of continued hydrocarbon development. [RAG: CLIMATE_RAG — IPCC AR6 Energy Systems Chapter 6]

Outlook

The structural configuration of Americas energy through 2030 rests on three interlocking dynamics. First, US tight oil production from the Permian Basin will remain the marginal swing supply for global oil markets, with production trajectory sensitive to prime acreage depletion rates and technological recovery advances rather than price signals alone. Second, the US LNG export position — now the world's largest — will deepen as additional liquefaction capacity commissioned in 2024-2026 reaches nameplate utilisation, tightening the linkage between Henry Hub pricing and global LNG netback values and cementing US geopolitical leverage across European and Asian import markets. Third, Brazil's Equatorial Margin drilling campaign, opening in 2026 with Petrobras's first wells in Foz do Amazonas and Pará-Maranhão, will determine whether the pre-salt growth arc can be extended into an entirely new geological domain, potentially adding a third major Americas supply growth vector alongside the Permian and deepwater Santos basin.

Against these production dynamics, the US policy environment under the January 2025 National Energy Emergency declaration has tilted sharply toward resource maximisation, creating uncertainty about the pace of IRA-funded clean energy deployment and the durability of US climate commitments. Mexico's Sonora Plan remains isolated as the only significant new renewable initiative in the country, indicating that hemispheric energy transition ambitions will be driven primarily by Brazil's wind and BECCS positioning rather than Mexican policy momentum. Canada's pipeline regulatory environment — shaped by persistent social opposition and infrastructure security concerns — continues to constrain the efficient integration of Alberta oil sands into continental and Atlantic export chains. The Americas energy order is consolidating around a US-Brazil axis of production dominance, with the geopolitical and climate consequences of that consolidation remaining deeply contested.

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